STAT 5243 · PROJECT 4 FINAL · COLUMBIA UNIVERSITY

Predicting the 2020-21 Premier League

End-to-end machine learning on 21 seasons of self-collected EPL data

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github.com/ZemingLiang/STAT5243-Project-4 · 2026-05-05

THE QUESTION

Can we predict the Premier League?

Three increasingly hard bars to clear:

Better than random. 33% accuracy floor.
Better than naive priors. Always-pick-Home ≈ 46%.
Better than the bookmakers. Sharps are the hardest baseline of all.

And we don't stop at one target.

Match outcome

Home / Draw / Away — 3-class probabilistic

Exact score

Bivariate Poisson via Dixon-Coles

Final standings

Monte Carlo simulation, 10k iterations

Beat the bookmaker

Kelly-fractional ROI vs Pinnacle close

Why football is genuinely hard

Tiny data

380 matches per season. Twenty-one seasons gives only ~7,890 rows — small for tabular ML.

High variance

One deflected goal can flip an outcome. Match-level noise is huge relative to signal.

Sharp bookmakers

Pinnacle's closing odds are the consensus of the most-informed actors with skin in the game.

Distribution shift

Covid 2020-21 played behind closed doors. Home-win rate dropped from ≈46% to ≈36%.

Plus: promoted teams have zero Premier League history → cold-start every August.

DATA

Five sources, all self-collected — no Kaggle

7,890
matches scraped

21
seasons (2000-01 → 2020-21)

5
independent sources

Source	Type	What we get
football-data.co.uk	Structured CSV	Results, half-time scores, closing odds
ClubElo.com	Structured API	Daily club Elo ratings, 2000+
FBref / soccerdata	Semi-structured	Advanced stats incl. xG (2017-18+)
Wikipedia	Semi-structured	Manager tenure, promotion/relegation
BBC + Guardian match reports	**Unstructured HTML**	**NLP → entities, sentiment, event tags**

Every source is harmonised into the same canonical team-name dictionary before any join — the UnknownTeamError pattern fails fast on new variants.

CLEANING SHOWCASE

Match-report HTML → feature row

1 Scrape
BBC / Guardian match-report HTML

→

2 Parse
BeautifulSoup → article text

→

3 NLP
spaCy NER + VADER + regex event classifier

→

4 Attach
Per-team sentiment + 12 feature columns

Worked example

INPUT · one sentence in

"Liverpool were sloppy in possession and were lucky to escape with a draw after a controversial VAR call denied Tottenham a stoppage-time winner."

OUTPUT · one feature row out

```
home_report_sentiment = -0.42
away_report_sentiment = +0.05
var_mention_count     = 1
controversy_flag      = 1
```

Three figures that shaped our modelling

FIGURE PLACEHOLDER

figures/eda_home_win_by_season.png

Home-win rate by season

Annotate the Covid-2020 dip from ~46% to ~36%.

FIGURE PLACEHOLDER

figures/eda_goal_distribution.png

Goals-per-match distribution

Right-skewed; mean ≈ 2.7 goals; supports the Poisson choice.

FIGURE PLACEHOLDER

figures/eda_team_style_pca.png

Team-style PCA, coloured by season

Style clusters drift over time — useful unsupervised feature.

KEY FINDING *The 2020-21 Covid season is a regime shift our model has to handle, not pretend away.*

FEATURE ENGINEERING

Six feature families, one strict cutoff rule

Form

Rolling N-match goals, points, xG. Leakage-safe shift.

Strength

ClubElo + own Elo update; home-Elo minus away-Elo.

Context

Rest days, derby flag, manager tenure.

Market

Pinnacle implied probabilities, vig-removed.

Unsupervised

PCA components + K-means style cluster ID.

NLP-derived

Per-team sentiment, event tags, controversy flag.

ANTI-LEAKAGE DISCIPLINE

No random k-fold. Walk-forward CV only. Every rolling feature uses a strict `as_of` cutoff: only matches with kickoff < current match contribute.

Unit-tested by `tests.test_no_temporal_leakage` *in the CI pipeline on every push.*

MODELS

Six models, one validation strategy

Model	Family	Why we included it
Always-Home	Naive baseline	Sanity floor — exposes log-loss penalty for over-confidence.
Class-Prior	Naive baseline	Predicts historical 46/25/29 H/D/A — the calibration floor.
Market-Implied	Strong baseline	Pinnacle closing odds, vig-removed. The hardest bar to clear.
Multinomial LR	Linear, interpretable	Reads off feature signs and magnitudes for sanity-checking.
Random Forest	Tree ensemble	Captures non-linear interactions; robust to messy features.
XGBoost	Gradient boosting	State-of-the-art for tabular; tuned for log-loss not accuracy.
Dixon-Coles	Bivariate Poisson	Domain-specific; gives joint score distribution + outcome.

VALIDATION

Walk-forward expanding-window CV · train 2000-01..2018-19, validate 2019-20, hold-out test 2020-21.
All hyperparameters tuned for log-loss (proper scoring rule), not raw accuracy.

RESULTS

The market wins. Our model gets within touching distance.

Model	Accuracy	Log-loss	Brier	F1 macro	AUC OvR
Market-implied (Pinnacle close)	0.516	0.997	0.592	0.385	0.669
Random Forest	0.513	1.017	0.606	0.461	0.646
Logistic regression	0.503	1.026	0.610	0.466	0.654
XGBoost	0.511	1.092	0.640	0.431	0.628
Class-prior baseline	0.379	1.099	0.669	0.183	0.500
Always-Home baseline	0.379	22.385	1.242	0.183	0.500

CONFUSION MATRIX

`figures/confusion_matrix_rf.png`

Most errors: Draw → Home / Away.
Draws are intrinsically the hardest class.

Random Forest is within 0.3pp accuracy and 0.02 log-loss of the market — using only the same engineered features the market presumably already prices.

And Always-Home's log-loss of 22.4? *That's the cost of being confidently wrong about Draws and Aways.*

Three bonus targets, one shared engine

Exact score

Dixon-Coles

Joint home-goals × away-goals from the bivariate Poisson model. Fitted with a 0.0065/day exponential decay weight; the tau correction term fixes under-prediction of (0,0), (0,1), (1,0), (1,1) scorelines.

Final standings

Monte Carlo

10,000 simulations of the 380-match season. Each iteration rolls every match's H/D/A from the per-match probabilities, accumulates points, ranks teams. Output: title prob, top-4 prob, relegation prob with 5/95 bands.

Beat the bookmaker

Kelly-fractional

Place when model edge = (p × decimal_odds − 1) > 0. Stake = Kelly fraction capped at 5% of bankroll. ROI on the 2020-21 season is the cleanest test — model conviction translated to dollars.

DEMO

Live: a Shiny app on Posit Connect Cloud

SCREENSHOT

figures/demo_1_predict_a_match.png

Tab 1. Predict-a-Match

Pick any 2020-21 fixture; see model probabilities, market probabilities, and the implied edge.

SCREENSHOT

figures/demo_2_season_simulator.png

Tab 2. Season Simulator

10,000 Monte Carlo runs of the 2020-21 table; live histogram of each team's finishing rank.

SCREENSHOT

figures/demo_3_leaderboard.png

Tab 3. Leaderboard

Full evaluation: log-loss, Brier, calibration plot, confusion matrix per model.

Live URL: `{TODO: pipeline owner to paste new Posit Connect Cloud URL}`

CONCLUSION

What we shipped, what's next, who did what

TL;DR

End-to-end ML on 7,890 self-collected matches. Six models, walk-forward CV, four interlinked targets, deployed Shiny app. Random Forest within 0.3pp accuracy of the bookmakers — but the bookmakers still win. The honest finding: the market is genuinely hard to beat, and that itself is the result.

FUTURE WORK

In-play features. Live xG, possession-by-half, lineup-strength deltas.

Hierarchical Bayes. Borrow strength across seasons; tighter cold-start for promoted clubs.

TEAM CONTRIBUTIONS

Member	Contribution
Zeming Liang	Coordinator; scrapers, models, app, deploy.
Xiying Chen	EDA, unsupervised features, NLP, report.
Collaborator	{TODO: confirm contribution}

Thank you. *Questions?* `github.com/ZemingLiang/STAT5243-Project-4`